

Five-Criterion TOPSIS and Full-Space Pareto Analysis

Saturated safety utility, weight sensitivity, and 9×9 dominance

The current matrix excludes success because it is constant and numerical peak acceleration because it is sampling-sensitive. Safety uses a bounded utility so that excessive separation is not rewarded without limit.

Study	9 methods × 4 scenarios × 10 paired repetitions
Navigation data	360 valid runs; 360 successful; 0 excluded; 0 failed
Source	Corrected manuscript supplied for this update (13 September 2026)

Agri-Social Navigation in Vineyard Corridors

1. Decision vector

$$X_i = [U_{\text{safe}}, T_p, T, N_{\text{stop}}, \sigma_v]$$

Unsafe is maximized; TP, T, Nstop, and σ_v are minimized. Nstop counts transitions into $v < 0.05$ m/s and can include the terminal stop, so it is not automatically interpreted as comfort. σ_v is comparable across all 360 metric files; low σ_v alone does not prove low acceleration or jerk.

2. Saturated safety utility

$$U_{\text{safe}}(d) = 0 \text{ if } d \leq 0.45 \text{ m; } (d - 0.45)/(1.20 - 0.45) \text{ if } 0.45 < d < 1.20 \text{ m; } 1 \text{ if } d \geq 1.20 \text{ m}$$

The transformation prevents distances beyond the 1.20 m personal boundary from receiving unlimited benefit. Spearman correlation is evaluated across all 360 runs; no pair exceeds $|\rho_s| \geq 0.70$, and the largest observed association is approximately 0.51 between Nstop and σ_v .

3. Weight profiles

Profile	Unsafe	TP	T	Nstop	σ_v	Interpretation
Equal	0.2000	0.2000	0.2000	0.2000	0.2000	Neutral reference
W1'	0.2667	0.2000	0.1333	0.2000	0.2000	Broad safety-oriented
W2'	0.2857	0.2597	0.0649	0.1299	0.2597	Safety, exposure, smoothness
W3'	0.2667	0.2000	0.0667	0.1333	0.3333	Strongest smoothness emphasis

The profiles are author-defined sensitivity scenarios; they are not expert-elicited or universal weights for harvesting, inspection, or transport.

4. Mean decision vectors

Method	Unsafe	TP [s]	T [s]	Nstop	σ_v [m/s]
M0	0.0381	14.624	96.277	1.000	0.1342
M1	0.1240	14.576	95.847	1.000	0.1231
M2	0.0515	15.017	95.576	1.000	0.1180
M3	0.0351	14.855	95.678	1.000	0.1201
M4	0.0685	16.077	95.642	1.000	0.1208
B1	0.3673	4.147	98.596	2.975	0.1806
B2	0.2182	15.836	93.661	1.050	0.0775
B3	0.1964	14.611	95.743	1.225	0.1201
B4	0.3488	4.978	97.338	1.000	0.0626

5. Closeness coefficients and ranks

Method	Equal CC	Rank	W1' CC	Rank	W2' CC	Rank	W3' CC	Rank
B4	0.955	1	0.953	1	0.950	1	0.955	1
B2	0.619	2	0.606	2	0.554	3	0.610	2
B3	0.557	3	0.542	4	0.478	4	0.502	4
B1	0.520	4	0.579	3	0.626	2	0.558	3
M1	0.508	5	0.465	5	0.380	5	0.411	5
M4	0.458	6	0.407	6	0.316	6	0.356	6
M2	0.455	7	0.399	7	0.312	7	0.353	7
M3	0.444	8	0.388	8	0.300	8	0.339	8
M0	0.437	9	0.382	9	0.285	9	0.314	9

B4 ranks first under equal weights and under $W1'$, $W2'$, and $W3'$. This characterizes this five-criterion formulation; it does not imply universal algorithmic superiority or replace task-specific weight selection.

6. Pareto dominance

Dominance is computed directly in the five-dimensional space: a method must be no worse in every criterion and strictly better in at least one. The 9×9 matrix contains one dominance relation: M1 dominates M0. The non-dominated set is {M1, M2, M3, M4, B1, B2, B3, B4}. The TP-versus-T plot is only a visual projection.

7. Reproducibility

- Preserve criterion order [Usafe, TP, T, Nstop, σv].
- Normalize without rounding before ideal-point distances are computed.
- Verify that every weight profile sums to one.
- Recompute Pareto dominance on full vectors, not the two-dimensional projection.